

# **SdCvP-X™: Electronic (Design) Verification Automation Platform**

**VERIPOINT TECHNOLOGIES PVT LTD**

AI Tools & Solutions Powering Semiconductors and Beyond

## Company Profile

**Veripoint Technologies** is a deep-tech product company addressing critical industry challenges, from automating semiconductor verification to delivering scalable, intelligent automation for industrial sectors through its AI-powered tools and engineering solutions.

### Industry Challenges

- **Semiconductors:** Manual verification slows development and increases the risk of costly silicon failures.
- **Industrial Sectors:** Scalable, intelligent automation solutions are still limited, hindering operations, efficiency and innovation.

### Solutions: Our Flagship Products & Premium DV Services

- **SdCvP-X™** : Proprietary AI-driven Electronic Design Verification Automation Platform.
- **AI-driven solutions** for manufacturing, automotive, telecommunications, healthcare and more.
- **Premium DV services** delivered by a team of experts with deep specialization in verifying complex designs.

**Founded:** Feb 2022 | **HQ:** Noida, UP, India | **Website:** [www.veripointtech.com](http://www.veripointtech.com) | **DPIIT:** DIPP158213

# SdCvP-X™: Electronic (Design) Verification Automation Platform

SdCvP-X™ is our proprietary AI tool that automates chip design verification, accelerating development and ensuring thorough verification.

## **Target Designs** (Standard & Custom)

- **IP, SoC, and Subsystem**

## **Specification (Design)**

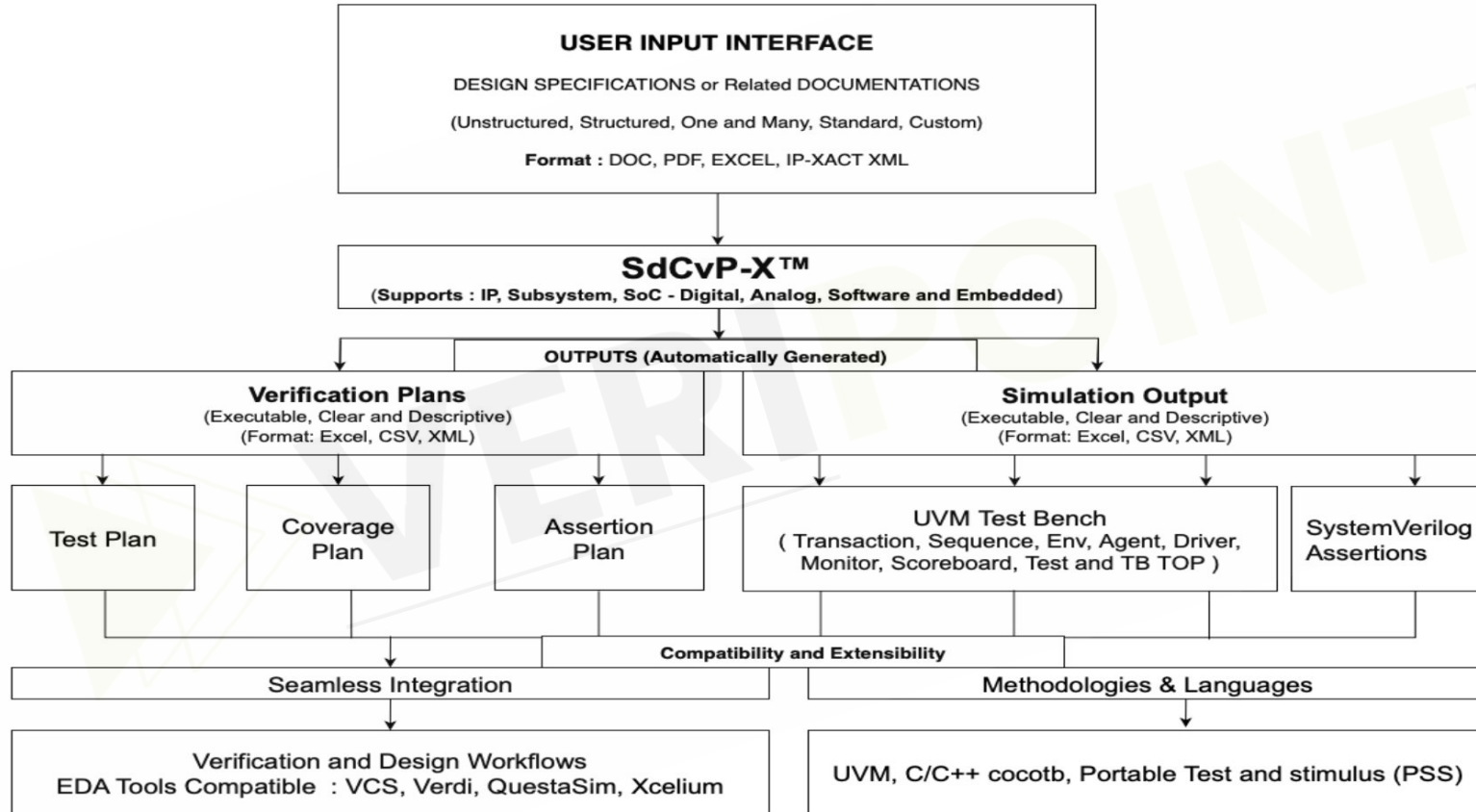
### **Benchmarks**

- High-speed and standard Interfaces like PCIe, CXL, NVMe, Ethernet, UCIe, USB, DisplayPort(DP, eDP), ARINC, and more
- Automotive & Networking Systems
- Analog IPs & Systems

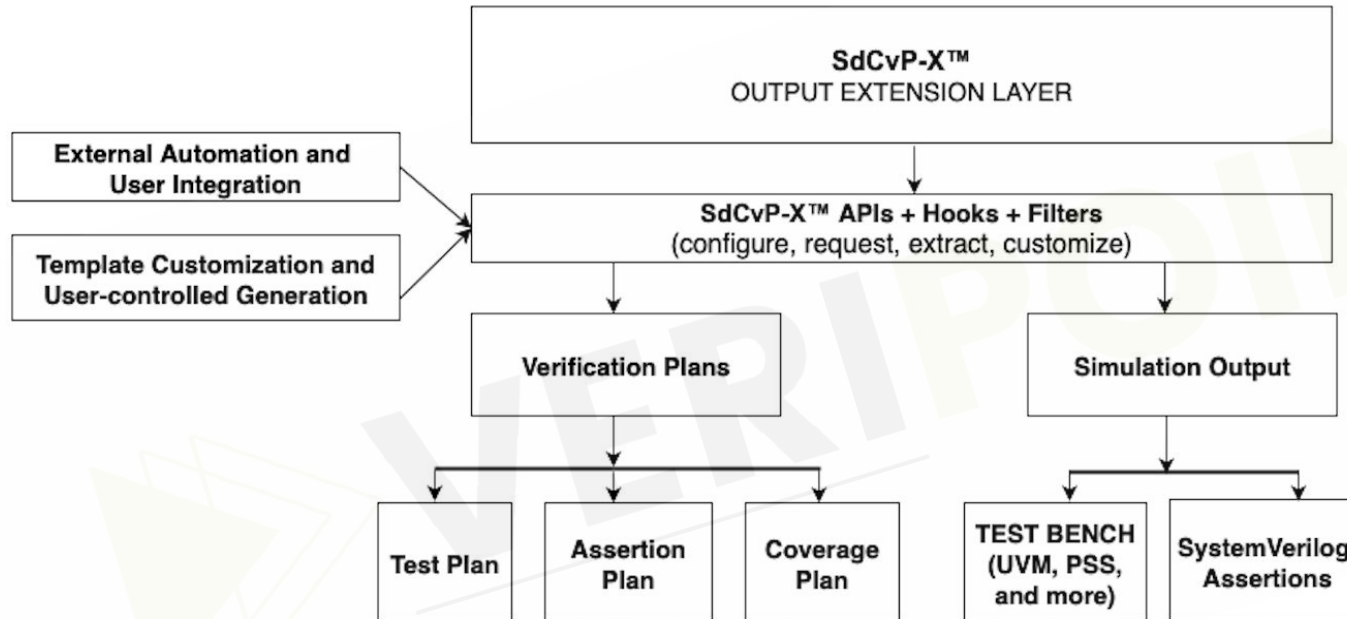
## **Key Features**

- **Auto-generates** features and executable verification plans - tests, assertions, and coverage directly from specifications in various formats including **IP-XACT XML**
- **Auto-generates** UVM Test Benches and SystemVerilog Assertions with custom format options directly from specifications
- **Dynamically adapts** to evolving specifications and syncs with manual plans
- **Auto-embeds** debug data for faster root-cause analysis
- **Integrates** with existing verification flows via Verification Manager & Planner of leading EDA tools
- **Extensible** through APIs for tool integration and Hooks/Plugins for customizable plans, testbenches, and SystemVerilog assertions

# SdCvP-X™: Capabilities & Verification Automation Workflow



# SdCvP-X™: Integration, Adaptation & Customization



# SdCvP-X™: Verifying Complex Designs

| Verification Challenges                                                                                                                                                                                                             | SdCvP-X™ Platform                                                                                                                                                                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Manual & fragmented planning, tests and assertions gaps                                                                                                                                                                             | Auto-generates Spec-aligned Verification Plans, UVM Test benches and SystemVerilog Assertions for Digital ( <b>IP, SoC, Subsystem</b> ) and Analog designs, directly from specifications including IP-XACT |
| Specification Alignment                                                                                                                                                                                                             | Plan evolves with specification + Version traceability                                                                                                                                                     |
| Manual Plan Alignment with Specification and upgrades                                                                                                                                                                               | Aligns manual plans to eliminate gaps and ensure consistency and boost productivity.                                                                                                                       |
| Debug Inefficiencies                                                                                                                                                                                                                | Traceability matrix + mapping to features/test points                                                                                                                                                      |
| Configuration Explosion                                                                                                                                                                                                             | Smart filtering, collapsing, and prioritization                                                                                                                                                            |
| Reuse across derivatives                                                                                                                                                                                                            | Scalable planning across standard & custom designs                                                                                                                                                         |
| <b><i>Unlike EDA simulators (VCS/Xcelium/Quest) that run testbenches, SdCvP-X™ automatically creates spec-aligned executable verification content, reducing manual effort by 50-60% and accelerating development by 30-40%.</i></b> |                                                                                                                                                                                                            |

## Who Benefits: Across the Project Hierarchy

| Role                          | Pain Point                                   | SdCvP-X™ Benefit                                                                                                                                                |
|-------------------------------|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DV Engineers                  | Long debug cycles, manual test list creation | Faster verification planning (test, coverage and assertion) + comprehensive test benches and system verilog assertions + structured debug : Boosts Productivity |
| Architects                    | Hard to validate spec implementation         | Full traceability from spec to test                                                                                                                             |
| Program/Project Managers      | Poor visibility on verification progress     | Actionable insights, milestone-based views                                                                                                                      |
| Business Leaders & Industries | Risk of re-spin, time-to-market delays       | Higher First Silicon Success, faster Time to market                                                                                                             |

## SdCvP-X™: Early Benchmarks Across Sectors and Designs

### **Data Centre & Storage (PCIe, CXL, USB, NVMe, Ethernet IPs & Subsystems)**

- Achieved *up to ~60% reduction* in verification planning and execution effort by auto-generating test, coverage, and assertion plans directly from specifications, ensuring spec-alignment.
- Achieved *~70%* faster testbench readiness by auto-generating UVM sequences, drivers, and assertions directly from evolving specs.

### **Automotive & Networking (UCIe, DisplayPort, ARINC, Automotive/Networking IPs, Subsystems & SoCs)**

- Achieved *~60%* reduction in rework effort by auto-syncing evolving specifications with verification plans, ensuring consistent traceability across derivatives.
- Achieved *~70%* faster verification ramp-up by reusing auto-generated verification features across protocol variants.

### **Mixed-Signal & Custom Systems (Custom Analog IPs & Mixed Systems)**

- Achieved *~60%* reduction in planning iterations by auto-aligning specifications with tests and assertions for analog and mixed-signal designs.



# SdCvP-X™: GUI & OUTPUT for CXL 3.1 IP (Eval)

**Design Domain**  
☒ IP ☐ SoC ☐ Subsystem

**Upload Specification**  
 Base Specification (PDF/docs)  
 CXL-3.1-Specification © 2019-2023 COMPUTE EXPRESS LINK CONSORTIUM, INC. ALL RIGHTS RESERVED.pdf  
 Base Register Specification (.xlsx/PDF)  
 Base Add-on Specification (PDF/docs)  
 Base Add-on Register Specification (.xlsx/PDF)

**Feature List**  
 Do you want specification feature list extraction? ☒ Yes ☐ No  
 Download Feature List

Do you want to upload modified feature list according to your design? ☐ Yes ☒ No

**Verification Plan Filters**  
 Configuration Enabled ☒ Yes ☐ No  
 Constraints Enabled ☒ Yes ☐ No  
 Coverage Enabled ☒ Yes ☐ No  
 Assertion Enabled ☒ Yes ☐ No  
 Debug Enabled ☒ Yes ☐ No  
 UVM - Sequence, Driver, Test Enabled ☒ Yes ☐ No  
 Confirm Filters  
 Generate Verification Plan

transcript\_cxl\_3\_1.txt

# Confidential Output – Generated by SdCvP-X, proprietary intellectual property of Veripoint Technologies Pvt Ltd.  
 # Generated from base specification: CXL-3.1-Specification © 2019-2023 COMPUTE EXPRESS LINK CONSORTIUM, INC. ALL RIGHTS RESERVED.pdf and no modification has been done in this file.  
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Domain selected: IP  
 Uploaded: CXL-3.1-Specification.pdf © 2019-2023 COMPUTE EXPRESS LINK CONSORTIUM, INC. ALL RIGHTS RESERVED  
 Reading Specification: CXL-3.1-Specification © 2019-2023 COMPUTE EXPRESS LINK CONSORTIUM, INC. ALL RIGHTS RESERVED.pdf  
 Created Processed Data: TEMP\_Data\_SdCvP-x from Specification CXL-3.1-Specification.pdf  
 Extracted 16 features.  
 Feature list saved : /Users/Veripoint\_Technologies/Desktop/sdcvp-x/feature\_list\_cxl\_3\_1.xlsx  
 Verification Plan Filters confirmed:  
 Configuration Enabled: Yes  
 Constraints Enabled: Yes  
 Coverage Enabled: Yes  
 Assertion Enabled: Yes  
 Debug Enabled: Yes  
 UVM - Sequence, Driver, Test Enabled: Yes  
 Verification plan generated and saved: /Users/Veripoint\_Technologies/Desktop/sdcvp-x/verification\_plan\_cxl\_3\_1.xlsx  
 Deleted Processed Data: TEMP\_Data\_SdCvP-x Securing Design Confidentiality

Verification Plan

Simulation Output

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| S.No. | Section | Layer/Component | Feature | DUT Type | Testpoint | Test Scenario | Procedure | Constraint | Assertion | Coverage | Spec Variables | Register Variables |
|-------|---------|-----------------|---------|----------|-----------|---------------|-----------|------------|-----------|----------|----------------|--------------------|
|-------|---------|-----------------|---------|----------|-----------|---------------|-----------|------------|-----------|----------|----------------|--------------------|

Verification Plan

Simulation Output

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| S.No. | Feature | Test Scenario | UVM Sequence | UVM Driver | UVM Test | SystemVerilog Assertions |
|-------|---------|---------------|--------------|------------|----------|--------------------------|
|-------|---------|---------------|--------------|------------|----------|--------------------------|

# SdCvP-X™: GUI & OUTPUT for Analog PM Bus System

Design Domain  
☒ IP ☐ SoC ☐ Subsystem

Upload Specification  
 Base Specification (PDF/docx)  
 Topic\_6\_Hesse.pdf  
 Base Register Specification (.xlsx/PDF)  
 Base Add-on Specification (PDF/docx)  
 Base Add-on Register Specification (.xlsx/PDF)

Feature List  
 Do you want specification feature list extraction?  
☒ Yes ☐ No  
 Download Feature List

Do you want to upload modified feature list according to your design?  
☐ Yes ☒ No

Verification Plan Filters  
 Configuration Enabled  
 Constraints Enabled  
 Coverage Enabled  
 Assertion Enabled  
 Debug Enabled  
 UVM - Sequence, Driver, Test Enabled

Confirm Filters  
 Generate Verification Plan

```
transcript_pm_bus.txt
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# Generated from base specification: Topic_6_Hesse.pdf and no modification has been done in this file.
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Domain selected: IP
Uploaded: Topic_6_Hesse.pdf
Reading Specification: Topic_6_Hesse.pdf
Created Processed Data: TEMP_Data_SdCvP-x from Specification Topic_6_Hesse.pdf
Extracted 33 features.
Feature list saved : /Users/Veripoint_Technologies/Desktop/sdcpv-x/feature_list_pm_bus.xlsx
Verification Plan Filters confirmed:
Configuration Enabled: Yes
Constraints Enabled: Yes
Coverage Enabled: Yes
Assertion Enabled: Yes
Debug Enabled: Yes
UVM - Sequence, Driver, Test Enabled: Yes
Verification plan generated and saved: /Users/Veripoint_Technologies/Desktop/sdcpv-x/verification_plan_pm_bus.xlsx
Deleted Processed Data: TEMP_Data_SdCvP-x Securing Design Confidentiality
```

## Verification Plan Simulation Output

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| S.No. | Section | Layer/Component | Feature | DUT Type | Testpoint | Test Scenario | Procedure | Constraint | Assertion | Coverage | Spec Variables | Register Variables |
|-------|---------|-----------------|---------|----------|-----------|---------------|-----------|------------|-----------|----------|----------------|--------------------|
|-------|---------|-----------------|---------|----------|-----------|---------------|-----------|------------|-----------|----------|----------------|--------------------|

## Verification Plan Simulation Output

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| S.No. | Feature | Test Scenario | UVM Sequence | UVM Driver | UVM Test | SystemVerilog Assertions |
|-------|---------|---------------|--------------|------------|----------|--------------------------|
|-------|---------|---------------|--------------|------------|----------|--------------------------|

# SdCvP-X™: Tiered Product Options

## Standard

- ✓ Comprehensive Verification Plan
- ✓ Features, Test Points & Scenarios
- ✓ Configuration & Constraints Plan

## Advanced

- ✓ All Standard Features
- ✓ Assertion Plan
- ✓ Enhanced Specification Evolution
- ✓ Testplan Alignment

Most Popular

## Premium

- ✓ All Advanced Features
- ✓ Coverage Plan
- ✓ Auto-Generated UVM Test Bench and SystemVerilog Assertions
- ✓ Spec Variables, Register Variables, Debug Info & Traceability
- ✓ Priority Support & Success Manager

## Contact Us

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